Topic VI: Land Planning and Development

Chapter

32. Subdivisions

33. Residential Planning

32 Subdivisions

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1. DEFINITION

The act of subdivision is the division of any tract or parcel of land into two or more parts for the purpose of sale or building development.

2. REGULATION OF SUBDIVISIONS

Unlike retracement surveys, surveys made for the purpose of the subdivision of land are creative in nature. Therefore, they are the original surveys for the subdivided land and will have a key role in the future ownership of the land being subdivided. Further, the care and imagination used in the planning of the subdivision process affect the entire community for many years to come. Creation of a subdivision involves more than furnishing a location for a home for a new member of a community and a profit for the developer. It requires planning for traffic, transportation, the location of schools, churches, and shopping centers, and the health and happiness of the citizens of the community.

There are many instances in the United States in the past where developers have created subdivisions that are a credit to their foresight and integrity without any regulatory laws. Yet there are also some poorly planned subdivisions that have caused communities to spend excessive amounts on street widening and resurfacing, reconstruction of sewer lines, establishment of additional drainage facilities, and increase in size of water mains. Such poor planning in the past has caused an acute awareness among state, county, and municipal officials of the need for the regulation of subdivision development. That awareness, together with increasing population and decreasing availability of land for development, have made it necessary for the states and the federal government to adopt laws regulating land divisions. Yet most subdivision regulation today is at the county level, and in some cases, the city level.

3. SUBDIVISION AND PLATTING LAWS

Subdivision law includes regulations for land use, types of streets and their dimensions, arrangement of lots and their sizes, land drainage, sewage disposal, protection of nature, and many other details. *Platting laws* include regulations for the recording of the subdivision plat, monumenting the parcels, establishing the accuracy of the survey, and means of identifying the parcels and their dimensions.

Most counties, and major cities, deal with developers of subdivisions through a planning commission. Therefore, the first step for a developer is to contact the planning commission of the designated approving agency for consultation. Many planning commissions require, at the outset, a data sheet that indicates the general features of the developer's plan and a location map that locates the proposed subdivision in relation to zoning regulation and existing community facilities.

4. SURVEYING FOR SUBDIVISIONS

The first step in the subdivision surveying process is a boundary survey. Unlike the interior monuments of a subdivision which, after platting, are correct and have no error of position, the exterior boundary monuments of a subdivision can be located in error. Establishing a subdivision does not take away the rights of the adjoiner. It is the surveyor's role to eliminate future boundary difficulties with a sound and defensible boundary survey before any detailed planning is made. Investigation as to any conflict with senior title holders should be made, and after certainty of location is established, the corners should be monumented.

The next step in the subdivision process is to establish a system of benchmarks for vertical control and prepare a topographic map. The map is essential in planning the subdivision, especially in regard to drainage and sanitary sewer plans. The topo survey provides the base for the preparation of a draft of the planned subdivision showing contours, street locations, and lots for preliminary approval by the planning commission.

The next step is preparation of a formal *preliminary plat*. That is actually the detailed plan for the subdivision and includes the names of the subdivider, engineer, or surveyor; a legal description of the tract; location and dimensions of all streets, lots, drainage structures, parks, and public areas; easement, lot, and block

numbers; contour lines; scale of map; north arrow; and date of preparation. After approval of the preliminary plat, the interior boundaries of the subdivision may be monumented and the developer can proceed with stake-out and construction operations.

5. FINAL PLAT

The *final plat* conforms to the preliminary plat, except for any changes imposed by the planning commission. It must be prepared and filed for record in accordance with the platting laws of the state. It establishes a legal description of the streets, residential lots, and other sites in the subdivision. This plat will serve as the basis for the description of the lots for formal conveyance.

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1. STORM DRAINAGE

The first step in planning subdivision drainage is a careful study of the contour map. Storm sewers and sanitary sewers are designed for gravity flow. Streets act as drainage collectors for storm runoff. Therefore, a general plan for storm sewers and sanitary sewers should be formulated before the streets are finally located.

Lots should drain to the street or to some open drainage system in the area. Lots should never receive drainage from the street. The Federal Housing Administration has set up requirements for lot drainage for various types of topography where this agency guarantees loans. These requirements call for the lot to slope away from the house for some distance in all directions. The ideal situation is for the lot to slope from the house to the street in the front of the lot and from the house to some drainage collector at the rear of the lot if one exists. Concrete alleyways with inverted crowns sometimes serve this purpose. Topography sometimes makes it necessary for the lot to slope from front to rear or from rear to front, but the house is always located so that the slope is away in all directions.

Storm water is carried along the gutter, which necessitates a curb. To avoid water collecting in pools, streets should have a minimum grade of 0.3% and preferably 0.5%. Water should not run across streets in valleys. Inlets should be planned for both sides of the streets at intersections.

The design of storm sewers depends on the amount of surface runoff to be carried in the storm sewers. The amount of rainfall that is absorbed by the surface soil depends on the perviousness of the soil, the intensity of the rainfall, the duration of the storm, and the slope of the surface. Water that is not absorbed is known as *surface runoff*. Prior to subdivision, farm or grazing land may absorb a large percentage of a given rainstorm. Streets, driveways, sidewalks, and rooftops, however, will cause most of the rain water to run off rather than be absorbed by the soil. The developer must be conscious of this fact in order to prevent damage to the development and to property downstream.

Factors included in the design of storm drains include area of drainage area in acres, shape of drainage area, slope of land in drainage area, use of land in drainage area (present and future), maximum intensity of rainfall, and frequency of maximum intensity.

2. SANITARY SEWERAGE

Septic tanks are undesirable in subdivisions and should not be used. The most desirable solution to sewage disposal is a collecting system connected to a municipal sewage disposal system. The second most desirable solution is a private collecting system and disposal plant.

Sanitary sewers flow by gravity, and their location depends on the topography of the land. They are often located under the streets, but they also may be located at the back of lots or in alleyways. The slope of the sewer should be such that the velocity will be between 2 ft/sec and 10 ft/sec. Manholes must be located at change of grade, junctions, and other points for inspection. Sewers must be laid in a straight line between manholes.

3. STREETS

Of major importance in the subdivision is the street system. Streets not only furnish an avenue for vehicle passage, they also furnish access to property for pedestrians, right-of-way for utility lines, channels for drainage, and access to fire plugs, garbage cans, and so on. The right street in the right place contributes to pleasant living. Insufficient street width creates traffic hazards, while excess width adds to the cost of construction and uses land that could be used for lots.

Local residential streets (also called *minor streets*) are designed to furnish access to private property. They are not designed to carry through traffic and should be designed to discourage it. Curved streets, loops, and cul-de-sacs all discourage speed. Long, curvilinear streets with block lengths up to 1800 ft have been found to be satisfactory. Where off-street parking is adequate, a roadway width of 26 ft (27 ft back-to-back of curbs) and a right-of-way of 50 ft are adequate for single-family residential neighborhoods. For multifamily neighborhoods, street widths should be 31 ft minimum (32 ft

back-to-back of curbs) with 60 ft right-of-way. A street width of 26 ft will not provide parking on both sides of the street with two lanes for traffic, but many homeowners find weaving in and out between parked cars slows traffic.

Collector streets carry traffic between local streets and arterial streets. They also furnish access to private property along the street. A width of 36 ft (37 ft back-to-back of curbs) will furnish two parking lanes and two traffic lanes. Right-of-way width should be 60 ft.

Arterial streets (also called *major streets*) move heavy traffic at relatively high speeds. Intersections are usually controlled by traffic lights. Many arterial streets provide six moving lanes with no parking. A 100 ft right-of-way will provide two 33 ft (back-to-back of curbs) lanes and a 14 ft median.

Cul-de-sacs are dead-end streets with turnarounds at the end. They are popular for single-family residences because of the privacy and freedom from noise they provide. They should not be more than 1000 ft in length because of the long turnaround. The turnaround circle should be not less than 40 ft in radius.

Loop streets have the same advantages as cul-de-sacs, plus the advantage of better circulation for fire trucks, delivery trucks, and police cars. Loop streets and cul-de-sacs can be used in odd corners of subdivisions.

4. BLOCKS

Until recent years, most subdivisions were laid out on a gridiron pattern. Modern subdivisions use curved streets with blocks 1400 ft to 1800 ft in length. These have proven to be more economical, using less street area by eliminating many cross-streets, and safer because of fewer intersections and slower speed on curved streets. They also provide more lots per undeveloped acre.

5. LOTS

Lot size should depend on the type of development, the topography of the land, and the expected cost per housing unit. Lot dimensions vary in different parts of the United States, but a minimum width of 60 ft and a minimum depth of 100 ft (or about 6000 ft^2) is considered desirable. Large, ranch-style houses with multicar garages opening on the front require more than 75 ft in width.

6. COVENANTS

To protect the interests of future property owners, developers often include certain restrictive *covenants* as a part of the deed to the property. These include such things as type of construction, minimum size and cost, setback distance, restrictions against advertising signs, raising of animals, parking of mobile homes, conducting certain types of commercial enterprises, and any other restrictions that will insure that the neighborhood is used for the purpose for which it was designed.

7. SETBACK LINES

To prevent locating buildings in such a way as to mar the general view of the neighborhood, restrictive covenants often designate the minimum distance from the front property line to the front of the building. To prevent monotony, staggered setback lines are sometimes used.

8. DENSITY ZONING

Many cities that have formerly controlled crowding in new neighborhoods by controlling the dimensions of lots have now adopted *density control*, which controls the maximum number of dwellings per acre. This has allowed more imagination in planning and better use of natural features.

9. CLUSTER PLANNING

A recent development in subdivision planning is the *cluster pattern*, where residences are clustered together in small, private sections of the subdivision with common open space. Cul-de-sacs and loop streets are adaptable to the cluster plan.